

LANs, WANs & WPANs

Three letters in front of "AN" decide how far a network reaches, how fast it runs, who owns it, and how exposed your data is. This dot point is about matching the right network type to the right job — and being able to defend that choice in an exam.

What this key knowledge covers

This summary distils everything you need for **KK05 — LANs, WANs & WPANs** in Cyber Security. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

01 The big idea: it's about scale, not cables

Students lose marks here by sorting networks into "wired" and "wireless." That's the wrong axis. **LAN, WAN and WPAN are sorted by geographic reach** — how much physical space the network covers. Cables vs radio is a separate question (and a LAN can be either).

Think about how far your **voice** carries. A **WPAN** is a whisper — only someone right next to you hears it (your earbuds, your watch). A **LAN** is talking across a room or a building — everyone in the house hears, no one outside does. A **WAN** is a phone call to another city — your words travel across infrastructure you don't own to reach someone far away. The technology differs, but what really changes is **how far it reaches**.

Picture the three scales as nested rings. Each ring out means more distance, more devices, more shared infrastructure — and a bigger surface for an attacker to reach.

02 LAN — Local Area Network one site

A **LAN** connects devices within a single physical location — a home, an office floor, a school, a hospital wing. One organisation (or person) owns and manages **all** the equipment: the switches, the cabling, the Wi-Fi access points, the local servers.

Clutch in real life

Clutch's **Carlton head office** is a textbook LAN. About 30 staff plug their desktops into wall sockets that run to a **switch** in a comms cupboard; a couple of **Wi-Fi access points** cover the meeting rooms; a **local file server** holds project files. Everything lives in one building and Clutch owns all of it, so an internal Figma export moves between two designers at full gigabit speed without ever touching the internet.

A LAN is like an **intercom system inside one building**. Press a button and your voice is instantly in the next room — fast, free, private. But the intercom stops at the front door. To reach the Geelong office you need a different system entirely: a phone line out (that's the WAN).

Wired or wireless? A LAN that uses Wi-Fi instead of cables is called a **WLAN** (Wireless LAN). It's still a LAN — same site, same owner — just delivered over radio. Don't confuse a WLAN with a WPAN: the WLAN covers the *building*; the WPAN covers the *person*.

03 WAN — Wide Area Network global

A **WAN** spans a large geographic area — between cities, states, or right around the globe. A WAN's job is to **join separate LANs together** so they act like one network. Crucially, no single organisation owns the whole path: you lease connectivity from telcos and your data crosses infrastructure other people control.

The internet is a **WAN** — in fact it's the world's largest one: a "network of networks" linking millions of LANs. Whenever an exam scenario mentions "over the internet," "to the cloud," or "between branches in different cities," you're looking at a WAN.

Clutch in real life

Clutch is a WAN business. Its app reaches **40,000 phones** spread across the whole state — and the only thing that can carry data that far is the internet (a WAN). The Carlton LAN talks to Clutch's **cloud servers** over a WAN link, and a small **Geelong support office** joins the company network across the same WAN. Because that traffic leaves the building and crosses networks Clutch doesn't own, it must be **encrypted** — which is exactly why KK04 covered VPNs.

A LAN is your driveway: short, private, yours. A WAN is the **public road system** — it'll take you anywhere in the country, but you share it with everyone, you don't own it, you pay tolls and rego, and it's slower and riskier than your own driveway. So you lock the car (encryption) before driving on it.

04 WPAN — Wireless Personal Area Network ~10 m

A **WPAN** is the smallest network type: a short-range wireless bubble — usually only a few metres — around **one person** and their devices. It's built for **low power and convenience**, not for speed or distance. The classic technologies are **Bluetooth**, **NFC** (tap-to-pay, ~4 cm) and **Zigbee** (smart-home sensors).

Clutch in real life

The WPAN lives **inside the learner's car**. The student's phone runs the Clutch app and pairs over **Bluetooth** to a hands-free car kit so route prompts play through the speakers; a paired **smartwatch** buzzes a gentle alert at hazard zones so the learner never looks down; an optional **OBD-II Bluetooth dongle** streams live speed to the app. All of that is one person's bubble of devices — a textbook WPAN — and none of it reaches past the car.

A WPAN is the **arm's-length bubble** around you. Your earbuds, watch and phone gossip among themselves, but the conversation can't be overheard from across the street. That tiny range is the whole point: it keeps power use and interference low, and it keeps the chatter personal.

Turn on your phone's **Bluetooth tethering** or pair it to a wireless speaker — you've just stood up a working WPAN. Tap your card on an EFTPOS terminal and that's NFC, a WPAN too. The study design literally suggests building one with a phone or an old wireless router to feel how short the range is.

05 Network Scale Lab

Pick a network type below and watch Clutch's data appear at that scale. The scorecard on the right shows the capabilities examiners reward — **range, speed, cost** and **devices** — so you can see how each one trades off against the others.

As you step **WPAN → LAN → WAN**, the range meter fills and the data travels further — but speed drops and cost climbs once traffic leaves the building. Flip on **Show attack surface** and watch how many more points an attacker can reach as the network widens. That growing exposure is the reason this dot point sits inside **Cyber Security**.

06 Capabilities side by side

This is the table to know cold. In a "compare" question, pick two or three of these rows and contrast them — never answer on a single dimension like speed alone.

Memory hook — read the meters left to right: **WPAN** = smallest everything; **LAN** = fastest, cheapest to run, one owner; **WAN** = biggest reach but slowest, priciest and most exposed.

08 Flip cards — lock in the definitions

A network covering a **single site** (building/campus), owned and managed by one organisation. High speed, low running cost. e.g. Clutch's Carlton office.

A network spanning a **large area** (cities → globe) that joins many LANs. No single owner; uses leased lines/the internet. e.g. Clutch's app reaching 40,000 phones.

A **short-range wireless** network (~10 m) around one person's devices. Bluetooth/NFC/Zigbee, low power. e.g. phone ↔ watch ↔ car kit.

WLAN = a wireless LAN (Wi-Fi) covering a **building**. **WPAN** = Bluetooth/NFC covering a **person** (~10 m). Different scale entirely.

A **WAN** — the largest one. It's a 'network of networks'. If a scenario says 'over the internet', it's a WAN.

The **LAN**. Data stays local over short cables (often 1–10 Gbps), so there's no carrier link or long-distance latency to slow it down.

The **WAN** — you pay ongoing fees to telcos for leased lines and data because you don't own the long-distance infrastructure.

Wider reach = bigger **attack surface**. WPAN is small/local; the WAN crosses untrusted networks, so data must be **encrypted in transit** (TLS/VPN).

09 Why this lives in Cyber Security

It can feel like a networking-trivia dot point, but it's in the **Cyber Security** outcome for a real reason: **scale is risk**. The wider a network reaches and the more shared infrastructure it crosses, the larger its **attack surface** — the number of places an attacker could get in.

Short range shrinks the threat, but an attacker **within metres** can still try bluesnarfing or hijack an unpaired device. Defence: secure pairing, PINs, turn Bluetooth off when unused.

One breached device sits **inside the trust boundary**, past the firewall . Defence: physical security, access control, segment the network, secure the Wi-Fi (WPA3).

Data crosses networks **nobody you trust** controls — open to interception in transit. Defence: encrypt everything end-to-end with **TLS and VPNs**.

Choosing a network type is also a **security decision**. Keeping sensitive data on the **LAN** limits exposure; the moment it must travel over the **WAN**, encryption stops being optional. Saying that out loud is what separates a top-band answer from a list of definitions.

10 Decode the command word

The command word tells you what the marker actually wants. Tap each to expand.

11 Quick quiz

Six multiple-choice checks. Pick an answer to see instant feedback — wrong picks explain the trap.

12 P·E·L writing trainer

Most network questions are worth 2–4 marks, and the structure that banks them is **P·E·L**: make your **Point**, give the **Evidence/Explanation**, then **Link** it back to the scenario. Write a response below and watch the keyword tracker light up as you hit the ideas a marker is looking for.

Name the network type and the capability you're using to judge it (range, speed, cost...).

Say *why* — give the technical reason that capability is true.

Tie it to the actual scenario and what the user needs.

P: Use a **WAN** — specifically the **mobile (4G/5G)** carrier network — for the car-to-internet link.

E: The cars are moving across the whole state, far beyond any single site, and only a WAN has the geographic **range** to reach head office from anywhere; mobile data is the only WAN access a moving car can use. **L:** Because that data crosses public infrastructure Clutch doesn't own, it must be **encrypted in transit**, so Clutch streams it over a secured/VPN connection to protect the learners' footage.

13 Exam practice — 5 questions · 20 marks

Each question shows a **weak** and a **strong** sample answer, then a marking guide. Mark your own attempt honestly — your score builds in the dock in the corner.

- 1 mark: short-range wireless network around one person (~10 m) / names Bluetooth-type tech
- 1 mark: a correct device example (earbuds, smartwatch, fitness band, car Bluetooth, NFC tap)
- 1 mark each (×2) for naming a relevant LAN capability — high speed, low cost, single ownership, easy security, no carrier fees
- 1 mark each (×2) for explaining WHY that capability follows from the LAN being local/one-site, tied to the office
- 1 mark: range contrast (single site vs global)
- 1 mark: speed contrast (LAN faster / local vs WAN slower / distance + shared links)

- 1 mark: cost contrast (LAN no carrier fees vs WAN ongoing fees)
- 1 mark: genuine comparative structure across all three (not three isolated statements)
- 1 mark: recommends a WAN / mobile (4G/5G) network
- 1 mark: justifies via range — cars move across the state, beyond any single site
- 1 mark: rules out alternatives (LAN/WLAN = one site, WPAN = ~10 m)
- 1 mark: notes the security implication of crossing untrusted networks (encrypt/VPN)
- 1-2 marks: explains that the three tasks sit at different scales (WPAN / LAN / WAN) with capabilities
- 1 mark: shows why one type can't serve all (e.g. WPAN can't span the state; WAN wasteful on one site)
- 1 mark: fairly notes the upside of the manager's view (simpler management/security)
- 1 mark: weighs both sides
- 1 mark: clear, justified judgement (reject the claim, match network to task)

Cover the answers, write your own response first, *then* reveal. Compare against the marking guide, award yourself marks, and tap your score. The weak answers aren't empty — they're the near-misses that lose marks, so learn to spot why.

15 One-screen cheat sheet

Digital Vector · VCE Applied Computing Units 1 & 2 · U2·O2 Cyber Security · KK05 — LANs, WANs & WPANs

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Distinguish · 3 marks

Distinguish between a **LAN**, a **WAN** and a **WPAN**, referring to the geographic area each covers.

✗ A weak answer

A LAN is local, a WAN is wide and a WPAN is wireless. (→ "Wireless" misnames WPAN's defining feature (it's personal/short-range) — 1-2.)

✓ A strong answer

A **LAN** (Local Area Network) connects devices within a single site such as a school or office. A **WAN** (Wide Area Network) connects networks across large distances — even between cities or countries — with the internet being the largest example. A **WPAN** (Wireless Personal Area Network) covers only a very short range around one person, such as a phone linked to wireless earbuds over Bluetooth.

How the marks are awarded

- 1 mark — LAN = single site/building.
- 1 mark — WAN = large/geographically dispersed.
- 1 mark — WPAN = very short, personal range.

Q2. Identify · 2 marks

Identify the most appropriate network type for (a) connecting a fitness tracker to a phone, and (b) linking a company's Melbourne and Sydney offices. Justify each.

✗ A weak answer

(a) LAN (b) LAN. (→ Both wrong; shows the categories aren't understood — 0.)

✓ A strong answer

(a) A **WPAN** — the tracker and phone are next to one person and communicate over a very short wireless range. (b) A **WAN** — Melbourne and Sydney are far apart, so a wide area network (or VPN over the internet) is needed to link the two LANs.

How the marks are awarded

- 1 mark — (a) WPAN with short-range justification.
- 1 mark — (b) WAN with distance justification.